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JEE MAIN PHYSICS DPP: UNIFORMLY ACCELERATED MOTION

Topic: Acceleration, Derivation of $v = u + at$ and $s = ut + \frac{1}{2}at^2$, Kinematic Equations | Time: 45 Min | Max Marks: 80

Marking Scheme: +4, -1 | Designed & Curated by: Syed Mazhar Ali

- Q1.** A particle starts from rest with constant acceleration a . The ratio of distances traversed by the particle in consecutive equal time intervals T ($s_1 : s_2 : s_3 : \dots$) follows:
- Natural number ratio: $1 : 2 : 3 : 4 : \dots$
 - Galileo's odd-number ratio: $1 : 3 : 5 : 7 : \dots$
 - Square ratio: $1 : 4 : 9 : 16 : \dots$
 - Even ratio: $2 : 4 : 6 : 8 : \dots$
- Q2.** In calculus, the fundamental derivation of $v = u + at$ and $s = ut + \frac{1}{2}at^2$ strictly requires which underlying physical condition?
- Velocity $v = \text{constant}$
 - Jerk $\frac{da}{dt} = \text{constant} \neq 0$
 - Acceleration $a = \text{constant}$
 - Acceleration $a \propto t$
- Q3.** A body moving with uniform acceleration covers 20 m in the 2nd second and 30 m in the 4th second of its motion. The distance covered in the 5th second is:
- 40 m
 - 45 m
 - 50 m
 - 35 m
- Q4.** A car moving at 20 m/s along a straight line can be stopped by applying brakes over a distance of 40 m. If the same car moves at 40 m/s, the minimum stopping distance under identical deceleration is:
- 160 m
 - 80 m
 - 120 m
 - 200 m
- Q5.** A particle starts from rest and accelerates uniformly for t seconds covering distance s_1 . It then decelerates at the same constant rate to rest in time t covering distance s_2 . The relation between s_1 and s_2 is:
- $s_1 = 2s_2$
 - $s_1 = s_2$
 - $s_2 = 2s_1$
 - $s_1 = 4s_2$
- Q6.** The displacement of a particle moving with uniform acceleration is given by $s = 3t + 4t^2$ m. The initial velocity u and acceleration a of the particle are:
- $u = 3$ m/s, $a = 4$ m/s²
 - $u = 6$ m/s, $a = 8$ m/s²
 - $u = 3$ m/s, $a = 8$ m/s²
 - $u = 4$ m/s, $a = 3$ m/s²
- Q7.** A point moves with uniform acceleration. If v_1, v_2 , and v_3 are the average velocities in three successive equal time intervals τ , the ratio $\frac{v_1 - v_2}{v_2 - v_3}$ is:
- 2
 - $\frac{1}{2}$
 - 3
 - 1
- Q8.** A train travelling at u brakes with constant deceleration a . The velocity of the train when it has covered half of its stopping distance is:
- $\frac{u}{\sqrt{2}}$
 - $\frac{u}{2}$
 - $\frac{u}{4}$
 - $\frac{\sqrt{3}u}{2}$
- Q9.** A particle starts from rest with uniform acceleration a . If the velocity after covering distance x is v , the distance covered when velocity becomes $2v$ is:
- $2x$
 - $4x$
 - $8x$
 - $16x$
- Q10.** Two cars A and B are moving in the same direction. Car A is moving with constant speed 10 m/s, while Car B starts from rest behind A with constant acceleration 2 m/s². If the initial separation is 25 m, the time taken by Car B to overtake Car A is:
- 10 s
 - 5 s
 - $5(1 + \sqrt{2})$ s
 - 2.5 s
- Q11.** A body starting from rest moves with uniform acceleration. If it covers distances x in first 2 s and y in next 2 s, then:
- $y = 2x$

- (b) $y = 4x$
 (c) $y = x$
 (d) $y = 3x$
- Q12.** A body travels with uniform acceleration a . It passes two points P and Q with speeds u and v respectively. The speed of the body at the midpoint of PQ is:
- (a) $\sqrt{\frac{u^2 + v^2}{2}}$
 (b) $\frac{u + v}{2}$
 (c) $\sqrt{uv}$
 (d) $\frac{2uv}{u + v}$
- Q13.** An object moves with uniform acceleration a starting with initial velocity u . The displacement during the n th second (s_n) is derived as:
- (a) $s_n = u + \frac{a}{2}(2n + 1)$
 (b) $s_n = u + \frac{a}{2}(2n - 1)$
 (c) $s_n = u + an$
 (d) $s_n = u + a(n - 1)$
- Q14.** A particle moves along a straight line with uniform acceleration $a = 2 \text{ m/s}^2$ and initial velocity $u = -6 \text{ m/s}$. The distance travelled by the particle in the first 4 s is:
- (a) 8 m
 (b) 18 m
 (c) 10 m
 (d) 12 m
- Q15.** A car accelerates from rest at rate α for some time, after which it decelerates at rate β to come to rest. If total time elapsed is T , the total distance covered is:
- (a) $\frac{\alpha\beta T^2}{\alpha + \beta}$
 (b) $\frac{(\alpha + \beta)T^2}{2\alpha\beta}$
 (c) $\frac{\alpha\beta T^2}{4(\alpha + \beta)}$
 (d) $\frac{\alpha\beta T^2}{2(\alpha + \beta)}$
- Q16.** A body released from rest from the top of a tower of height H falls under gravity ($g = \text{constant}$). The distance fallen in the last second of its motion is equal to the distance fallen in the first 3 seconds. The height H is: (Take $g = 10 \text{ m/s}^2$)
- (a) 125 m
 (b) 80 m
 (c) 45 m
 (d) 100 m
- Q17.** The displacement x of a particle moving with uniform acceleration is related to velocity v as $v = \sqrt{49 + 14x}$. The initial velocity and acceleration of the particle are:
- (a) 7 m/s, 14 m/s^2
 (b) 7 m/s, 7 m/s^2
 (c) 49 m/s, 7 m/s^2
 (d) 0 m/s, 14 m/s^2
- Q18.** A driver traveling at 72 km/h sees an obstruction on the road and applies brakes after a reaction time of 0.5 s. If the brakes cause a uniform deceleration of 5 m/s^2 , the total distance travelled before the car stops is:
- (a) 40 m
 (b) 30 m
 (c) 50 m
 (d) 60 m
- Q19.** A bullet fired into a target loses half of its velocity after penetrating 3 cm. How much further will it penetrate before coming to rest, assuming uniform retardation?
- (a) 1.5 cm
 (b) 2.0 cm
 (c) 0.5 cm
 (d) 1.0 cm
- Q20.** A body moves in a straight line with uniform acceleration. If it covers distances s_1 in time t_1 and s_2 in subsequent time t_2 , the uniform acceleration a is given by:
- (a) $\frac{2(s_2t_1 - s_1t_2)}{t_1t_2(t_1 + t_2)}$
 (b) $\frac{s_2t_1 - s_1t_2}{t_1t_2(t_1 + t_2)}$
 (c) $\frac{2(s_1t_2 - s_2t_1)}{t_1t_2(t_1 + t_2)}$
 (d) $\frac{s_2 - s_1}{t_1 + t_2}$

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(b)	6	(c)	11	(d)	16	(a)
2	(c)	7	(d)	12	(a)	17	(b)
3	(d)	8	(a)	13	(b)	18	(c)
4	(a)	9	(b)	14	(c)	19	(d)
5	(b)	10	(c)	15	(d)	20	(a)

Detailed Solutions

Sol 1. Correct Option: (b)

For a body starting from rest ($u = 0$) with constant acceleration a :

$$s(t) = \frac{1}{2}at^2.$$

$$\bullet s_1 = s(T) = \frac{1}{2}aT^2$$

$$\bullet s_2 = s(2T) - s(T) = 3 \left(\frac{1}{2}aT^2 \right)$$

$$\bullet s_3 = s(3T) - s(2T) = 5 \left(\frac{1}{2}aT^2 \right).$$

Thus, $s_1 : s_2 : s_3 : \dots = 1 : 3 : 5 : 7 : \dots$ (Galileo's odd-number ratio).

Sol 2. Correct Option: (c)

$$\bullet \text{ From } a = \frac{dv}{dt} \implies dv = a dt.$$

$$\text{Integrating: } \int_u^v dv = a \int_0^t dt \implies v = u + at \text{ (valid only when } a \text{ is constant).}$$

$$\bullet \text{ Since } v = \frac{ds}{dt} = u + at \implies ds = (u + at)dt.$$

$$\text{Integrating: } s = \int_0^t (u + at)dt = ut + \frac{1}{2}at^2.$$

Sol 3. Correct Option: (d)

$$s_n = u + \frac{a}{2}(2n - 1).$$

$$s_2 = u + \frac{a}{2}(3) = 20 \implies u + 1.5a = 20$$

$$s_4 = u + \frac{a}{2}(7) = 30 \implies u + 3.5a = 30$$

Subtracting gives $2a = 10 \implies a = 5 \text{ m/s}^2$ and $u = 12.5 \text{ m/s}$.

For $n = 5$:

$$s_5 = 12.5 + \frac{5}{2}(9) = 12.5 + 22.5 = 35 \text{ m}$$

Sol 4. Correct Option: (a)

Using $v^2 = u^2 - 2as$ with final velocity $v = 0$:

$$\text{Stopping distance } s = \frac{u^2}{2a} \propto u^2.$$

When speed is doubled ($u \rightarrow 2u$):

$$s' = 4s = 4 \times 40 \text{ m} = 160 \text{ m}$$

Sol 5. Correct Option: (b)

$$\bullet \text{ Acceleration phase: } s_1 = \frac{1}{2}at^2, \text{ achieving peak}$$

speed $v = at$.

$$\bullet \text{ Deceleration phase: covers distance } s_2 = vt - \frac{1}{2}at^2 = (at)t - \frac{1}{2}at^2 = \frac{1}{2}at^2.$$

Therefore, $s_1 = s_2$.

Sol 6. Correct Option: (c)

$$\text{Given } s = 3t + 4t^2.$$

Comparing with standard kinematic equation

$$s = ut + \frac{1}{2}at^2:$$

$$u = 3 \text{ m/s}$$

$$\frac{1}{2}a = 4 \implies a = 8 \text{ m/s}^2.$$

Sol 7. Correct Option: (d)

Let velocities at boundaries be u_0, u_1, u_2, u_3 with interval τ :

$$u_1 = u_0 + a\tau, u_2 = u_0 + 2a\tau, u_3 = u_0 + 3a\tau.$$

Average velocities:

$$v_1 = \frac{u_0 + u_1}{2} = u_0 + \frac{1}{2}a\tau$$

$$v_2 = \frac{u_1 + u_2}{2} = u_0 + \frac{3}{2}a\tau$$

$$v_3 = \frac{u_2 + u_3}{2} = u_0 + \frac{5}{2}a\tau$$

$$v_1 - v_2 = -a\tau, \text{ and } v_2 - v_3 = -a\tau.$$

$$\text{Ratio: } \frac{v_1 - v_2}{v_2 - v_3} = \frac{-a\tau}{-a\tau} = 1.$$

Sol 8. Correct Option: (a)

$$\text{Total stopping distance } S = \frac{u^2}{2a}.$$

$$\text{At half distance } s = S/2 = \frac{u^2}{4a}:$$

$$v^2 = u^2 - 2a \left(\frac{u^2}{4a} \right) = u^2 - \frac{u^2}{2} = \frac{u^2}{2}$$

$$v = \frac{u}{\sqrt{2}}$$

Sol 9. Correct Option: (b)

$$\text{Using } v^2 = 2ax \implies x \propto v^2.$$

When velocity increases from v to $2v$:

$$x' = \frac{(2v)^2}{2a} = 4 \left(\frac{v^2}{2a} \right) = 4x$$

Sol 10. Correct Option: (c)

In relative frame of car A:

$$u_{\text{rel}} = u_B - u_A = 0 - 10 = -10 \text{ m/s.}$$

$$a_{\text{rel}} = a_B - a_A = 2 - 0 = 2 \text{ m/s}^2.$$

Relative displacement to overtake: $s_{\text{rel}} = 25 \text{ m.}$

$$s_{\text{rel}} = u_{\text{rel}}t + \frac{1}{2}a_{\text{rel}}t^2 \Rightarrow 25 = -10t + \frac{1}{2}(2)t^2$$

$$t^2 - 10t - 25 = 0$$

$$t = \frac{10 + \sqrt{100 + 100}}{2} = 5(1 + \sqrt{2}) \text{ s}$$

Sol 11. Correct Option: (d)

$$s(t) = \frac{1}{2}at^2.$$

$$x = s(2) = \frac{1}{2}a(2)^2 = 2a.$$

$$s(4) = \frac{1}{2}a(4)^2 = 8a.$$

$$y = s(4) - s(2) = 8a - 2a = 6a.$$

Therefore, $y = 3x$.

Sol 12. Correct Option: (a)

Let total distance between P and Q be $2s$.

$$v^2 = u^2 + 2a(2s) \Rightarrow 2as = \frac{v^2 - u^2}{2}.$$

For velocity v_m at midpoint (distance s):

$$v_m^2 = u^2 + 2as = u^2 + \frac{v^2 - u^2}{2} = \frac{u^2 + v^2}{2}$$

$$v_m = \sqrt{\frac{u^2 + v^2}{2}}$$

Sol 13. Correct Option: (b)

Displacement in n seconds: $s(n) = un + \frac{1}{2}an^2$.

Displacement in $(n - 1)$ seconds: $s(n - 1) = u(n - 1) + \frac{1}{2}a(n - 1)^2$.

$$s_n = s(n) - s(n - 1)$$

$$= u[n - (n - 1)] + \frac{1}{2}a[n^2 - (n^2 - 2n + 1)]$$

$$= u + \frac{a}{2}(2n - 1)$$

Sol 14. Correct Option: (c)

Velocity $v(t) = u + at = -6 + 2t$.

Particle turns at $v(t) = 0 \Rightarrow t = 3 \text{ s.}$

$$\bullet s(3) = u(3) + \frac{1}{2}a(3)^2 = -6(3) + \frac{1}{2}(2)(9) = -9 \text{ m.}$$

$$\bullet s(4) = -6(4) + \frac{1}{2}(2)(16) = -8 \text{ m.}$$

Total distance = $|s(3) - s(0)| + |s(4) - s(3)| = |-9| + |-8 - (-9)| = 9 + 1 = 10 \text{ m.}$

Sol 15. Correct Option: (d)

Peak velocity $v_{\text{max}} = \alpha t_1 = \beta t_2$.

$$T = t_1 + t_2 = \frac{v_{\text{max}}}{\alpha} + \frac{v_{\text{max}}}{\beta} = v_{\text{max}} \left(\frac{\alpha + \beta}{\alpha\beta} \right) \Rightarrow$$

$$v_{\text{max}} = \frac{\alpha\beta T}{\alpha + \beta}.$$

Total distance from v - t graph area:

$$s = \frac{1}{2}Tv_{\text{max}} = \frac{1}{2}T \left(\frac{\alpha\beta T}{\alpha + \beta} \right) = \frac{\alpha\beta T^2}{2(\alpha + \beta)}$$

Sol 16. Correct Option: (a)

Total time to fall height H : $T = \sqrt{2H/g}$.

$$\text{Distance in first 3 s: } s_3 = \frac{1}{2}g(3)^2 = 45 \text{ m.}$$

Distance in last second:

$$s_T = g \left(T - \frac{1}{2} \right) = 45 \Rightarrow 10(T - 0.5) = 45$$

$$T - 0.5 = 4.5 \Rightarrow T = 5 \text{ s}$$

$$\text{Total height } H = \frac{1}{2}gT^2 = \frac{1}{2}(10)(5)^2 = 125 \text{ m.}$$

Sol 17. Correct Option: (b)

$$\text{Squaring } v = \sqrt{49 + 14x} \Rightarrow v^2 = 49 + 14x.$$

Comparing with $v^2 = u^2 + 2ax$:

$$u^2 = 49 \Rightarrow u = 7 \text{ m/s}$$

$$2a = 14 \Rightarrow a = 7 \text{ m/s}^2.$$

Sol 18. Correct Option: (c)

$$\text{Speed } u = 72 \text{ km/h} = 72 \times \frac{5}{18} = 20 \text{ m/s.}$$

$$\bullet \text{ Distance during reaction time: } s_1 = ut_r = 20 \times 0.5 = 10 \text{ m.}$$

$$\bullet \text{ Stopping distance under braking: } s_2 = \frac{u^2}{2a} =$$

$$\frac{(20)^2}{2(5)} = \frac{400}{10} = 40 \text{ m.}$$

$$\text{Total stopping distance} = s_1 + s_2 = 10 + 40 = 50 \text{ m.}$$

Sol 19. Correct Option: (d)

Initial speed u . After $s_1 = 3 \text{ cm}$, speed is $u/2$:

$$(u/2)^2 = u^2 - 2a(3) \Rightarrow \frac{u^2}{4} = u^2 - 6a \Rightarrow 6a =$$

$$\frac{3}{4}u^2 \Rightarrow 2a = \frac{u^2}{4}.$$

Let further distance penetrated be s_2 until $v = 0$:

$$0 = (u/2)^2 - 2as_2 \Rightarrow \frac{u^2}{4} = 2as_2.$$

$$\text{Substitute } 2a = \frac{u^2}{4}:$$

$$\frac{u^2}{4} = \left(\frac{u^2}{4} \right) s_2 \Rightarrow s_2 = 1.0 \text{ cm}$$

Sol 20. Correct Option: (a)

$$s_1 = ut_1 + \frac{1}{2}at_1^2 \Rightarrow \frac{s_1}{t_1} = u + \frac{1}{2}at_1.$$

In total time $(t_1 + t_2)$, displacement is $s_1 + s_2$:

$$\frac{s_1 + s_2}{t_1 + t_2} = u + \frac{1}{2}a(t_1 + t_2).$$

Subtracting the two equations gives:

$$\frac{s_1 + s_2}{t_1 + t_2} - \frac{s_1}{t_1} = \frac{1}{2}at_2$$

$$\frac{s_2t_1 - s_1t_2}{t_1(t_1 + t_2)} = \frac{1}{2}at_2 \Rightarrow a = \frac{2(s_2t_1 - s_1t_2)}{t_1t_2(t_1 + t_2)}$$

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